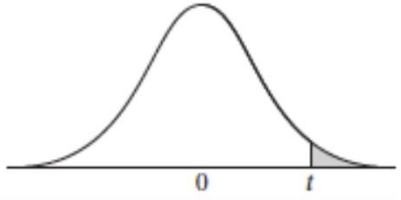
Chapter 6

Confidence intervals

- From Central Limit Theorem:
  - $\frac{\bar{Y}-\mu}{\sigma/\sqrt{n}}$ has a N(0,1) distribution
- Using our tables:
  - $P\left(-1.96 < \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} < 1.96\right) \approx .95$
- Isolating μ:
  - $P(\bar{Y} - (1.96 \times \sigma/\sqrt{n}) < \mu < \bar{Y} + (1.96 \times \sigma/\sqrt{n})) \approx .95$
- μ, the population mean, is *not* random
- The sample and confidence interval are random
- But we don't know σ

t-distribution

- If data is normally distributed, we can use it
- As df → ∞, t-distribution goes to normal distribution
- Each t value depends on df and confidence level
- Critical value – odds of getting a value this or more extreme
- Two-sided:
  - 90% confidence: $t_{.05}$
  - 99% confidence: $t_{.005}$

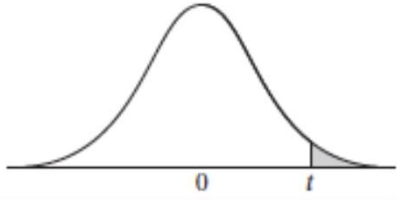


Critical Values of Student's *t* Distribution

	UPPER TAIL PROBABILITY									
df	0.20	0.10	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587
11	0.876	1.363	1.796	1.928	2.096	2.201	2.328	2.718	3.106	4.437
12	0.873	1.356	1.782	1.912	2.076	2.179	2.303	2.681	3.055	4.318
13	0.870	1.350	1.771	1.899	2.060	2.160	2.282	2.650	3.012	4.221
14	0.868	1.345	1.761	1.888	2.046	2.145	2.264	2.624	2.977	4.140
15	0.866	1.341	1.753	1.878	2.034	2.131	2.249	2.602	2.947	4.073
16	0.865	1.337	1.746	1.869	2.024	2.120	2.235	2.583	2.921	4.015
17	0.863	1.333	1.740	1.862	2.015	2.110	2.224	2.567	2.898	3.965
18	0.862	1.330	1.734	1.855	2.007	2.101	2.214	2.552	2.878	3.922
19	0.861	1.328	1.729	1.850	2.000	2.093	2.205	2.539	2.861	3.883
20	0.860	1.325	1.725	1.844	1.994	2.086	2.197	2.528	2.845	3.850
21	0.859	1.323	1.721	1.840	1.988	2.080	2.189	2.518	2.831	3.819
22	0.858	1.321	1.717	1.835	1.983	2.074	2.183	2.508	2.819	3.792
23	0.858	1.319	1.714	1.832	1.978	2.069	2.177	2.500	2.807	3.768
24	0.857	1.318	1.711	1.828	1.974	2.064	2.172	2.492	2.797	3.745
25	0.856	1.316	1.708	1.825	1.970	2.060	2.167	2.485	2.787	3.725
26	0.856	1.315	1.706	1.822	1.967	2.056	2.162	2.479	2.779	3.707
27	0.855	1.314	1.703	1.819	1.963	2.052	2.158	2.473	2.771	3.690
28	0.855	1.313	1.701	1.817	1.960	2.048	2.154	2.467	2.763	3.674
29	0.854	1.311	1.699	1.814	1.957	2.045	2.150	2.462	2.756	3.659
30	0.854	1.310	1.697	1.812	1.955	2.042	2.147	2.457	2.750	3.646
40	0.851	1.303	1.684	1.796	1.936	2.021	2.123	2.423	2.704	3.551
50	0.849	1.299	1.676	1.787	1.924	2.009	2.109	2.403	2.678	3.496
60	0.848	1.296	1.671	1.781	1.917	2.000	2.099	2.390	2.660	3.460
70	0.847	1.294	1.667	1.776	1.912	1.994	2.093	2.381	2.648	3.435
80	0.846	1.292	1.664	1.773	1.908	1.990	2.088	2.374	2.639	3.416
100	0.845	1.290	1.660	1.769	1.902	1.984	2.081	2.364	2.626	3.390
140	0.844	1.288	1.656	1.763	1.896	1.977	2.073	2.353	2.611	3.361
1000	0.842	1.282	1.646	1.752	1.883	1.962	2.056	2.330	2.581	3.300
∞	0.842	1.282	1.645	1.751	1.881	1.960	2.054	2.326	2.576	3.291
	60%	80%	90%	92%	94%	95%	96%	98%	99%	99.9%
	CRITICAL VALUE FOR CONFIDENCE LEVEL									

t-distribution

$\bar{Y} \pm (t \times s/\sqrt{n})$ where df = n-1



Critical Values of Student's *t* Distribution

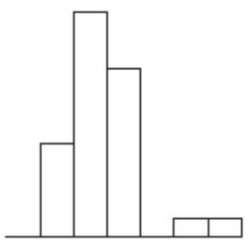
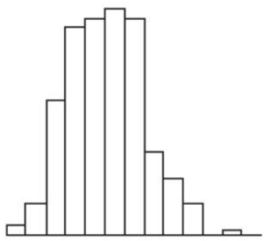
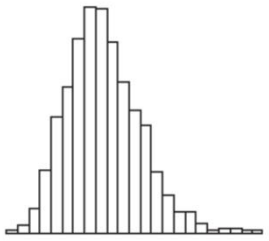
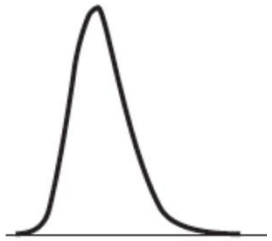
df	UPPER TAIL PROBABILITY									
	0.20	0.10	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587
11	0.876	1.363	1.796	1.928	2.096	2.201	2.328	2.718	3.106	4.437
12	0.873	1.356	1.782	1.912	2.076	2.179	2.303	2.681	3.055	4.318
13	0.870	1.350	1.771	1.899	2.060	2.160	2.282	2.650	3.012	4.221
14	0.868	1.345	1.761	1.888	2.046	2.145	2.264	2.624	2.977	4.140
15	0.866	1.341	1.753	1.878	2.034	2.131	2.249	2.602	2.947	4.073
16	0.865	1.337	1.746	1.869	2.024	2.120	2.235	2.583	2.921	4.015
17	0.863	1.333	1.740	1.862	2.015	2.110	2.224	2.567	2.898	3.965
18	0.862	1.330	1.734	1.855	2.007	2.101	2.214	2.552	2.878	3.922
19	0.861	1.328	1.729	1.850	2.000	2.093	2.205	2.539	2.861	3.883
20	0.860	1.325	1.725	1.844	1.994	2.086	2.197	2.528	2.845	3.850
21	0.859	1.323	1.721	1.840	1.988	2.080	2.189	2.518	2.831	3.819
22	0.858	1.321	1.717	1.835	1.983	2.074	2.183	2.508	2.819	3.792
23	0.858	1.319	1.714	1.832	1.978	2.069	2.177	2.500	2.807	3.768
24	0.857	1.318	1.711	1.828	1.974	2.064	2.172	2.492	2.797	3.745
25	0.856	1.316	1.708	1.825	1.970	2.060	2.167	2.485	2.787	3.725
26	0.856	1.315	1.706	1.822	1.967	2.056	2.162	2.479	2.779	3.707
27	0.855	1.314	1.703	1.819	1.963	2.052	2.158	2.473	2.771	3.690
28	0.855	1.313	1.701	1.817	1.960	2.048	2.154	2.467	2.763	3.674
29	0.854	1.311	1.699	1.814	1.957	2.045	2.150	2.462	2.756	3.659
30	0.854	1.310	1.697	1.812	1.955	2.042	2.147	2.457	2.750	3.646
40	0.851	1.303	1.684	1.796	1.936	2.021	2.123	2.423	2.704	3.551
50	0.849	1.299	1.676	1.787	1.924	2.009	2.109	2.403	2.678	3.496
60	0.848	1.296	1.671	1.781	1.917	2.000	2.099	2.390	2.660	3.460
70	0.847	1.294	1.667	1.776	1.912	1.994	2.093	2.381	2.648	3.435
80	0.846	1.292	1.664	1.773	1.908	1.990	2.088	2.374	2.639	3.416
100	0.845	1.290	1.660	1.769	1.902	1.984	2.081	2.364	2.626	3.390
140	0.844	1.288	1.656	1.763	1.896	1.977	2.073	2.353	2.611	3.361
1000	0.842	1.282	1.646	1.752	1.883	1.962	2.056	2.330	2.581	3.300
∞	0.842	1.282	1.645	1.751	1.881	1.960	2.054	2.326	2.576	3.291
	60%	80%	90%	92%	94%	95%	96%	98%	99%	99.9%
	CRITICAL VALUE FOR CONFIDENCE LEVEL									

Confidence interval

- A higher confidence interval is wider
  - 95% confidence interval is wider than 90%
  - 100% confidence interval would include everything
- Confidence interval is based on sample
- Sample is random so the confidence interval is random
  - Confidence interval only works if sample is from a simple random sample
  - If the sample size is small, the distribution **must** be approximately normal
  - If sample size is large (n ~ 30), distribution does not need to be normal (CLT)
    - But look for outliers regardless
- 95% confidence interval means if we repeat this many times, the population mean would be included in 95% of them
  - A particular calculated interval either has the true population mean or not
- As sample size increases, the width of the interval decreases

Standard error

- $SE_{\bar{Y}} = \frac{s}{\sqrt{n}}$
  - E.g. to lower SE by a factor of 2, we need 4 times the number of samples
- Measures **uncertainty** in our **estimate** of the population mean (μ)
- Different than standard deviation (SD) which measures the dispersion of the data

	$n = 28$	$n = 280$	$n = 2{,}800$	$n \to \infty$
$\bar{y}$	5.17	5.19	5.14	$\bar{y} \to \mu$
s	0.65	0.67	0.65	$s \to \sigma$
SE	0.12	0.040	0.012	SE $\to 0$
Sample distribution				

One-sided confidence intervals

- t
  - We don’t need the factor of 2
  - E.g. for a 90% confidence interval we use $t_{0.1}$
- Lower bound:
  - $\bar{Y} - (t \times s/\sqrt{n})$
  - Confidence interval: $(\bar{Y} - (t \times s/\sqrt{n}), \infty)$
- Upper bound:
  - $\bar{Y} + (t \times s/\sqrt{n})$
  - Confidence interval:$(-\infty, \bar{Y} + (t \times s/\sqrt{n}))$

Comparing two means

- Requires 2 independent samples
- We can compare μ_1 and μ_2:
  - $\mu_1 - \mu_2$, is it different than 0?
  - This is the same as $\mu_1 = \mu_2$
- Standard error of each distribution:
  - $SE_{\bar{Y}_1} = \frac{s_1}{\sqrt{n_1}}$
  - $SE_{\bar{Y}_2} = \frac{s_2}{\sqrt{n_2}}$
- Standard error of the difference:
  - $SE_{\bar{Y}_1 - \bar{Y}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Confidence interval for $\mu_1 - \mu_2$

- Requires 2 samples to be independent and normal
  - $(\bar{y}_1 - \bar{y}_2) \pm (t \times SE_{(\bar{Y}_1 - \bar{Y}_2)})$
- Degrees of freedom
  - $df = \frac{(SE_1^2 + SE_2^2)^2}{SE_1^4/(n_1 - 1) + SE_2^4/(n_2 - 1)}$
- For textbook problems, it will be computed for you
- R can compute this automatically
- Otherwise, we can use 2 approximations:
  - **Conservative approximation: min((n_1-1), (n_2-1))**
    - **Use this one, for calculated df round down**
  - Liberal approximation: n_1+n_2-2